Data Loading

Shape, dtypes & target distribution

```
Shape: (45211, 16)
Missing values: 0 across all columns
Target: No (0): 39,922 (88.3%)  |  Yes (1): 5,289 (11.7%)
Class ratio: 7.5:1
```

45,211 rows, 16 columns, zero missing values. The 7.5:1 class imbalance is the defining modelling challenge — it rules out accuracy as a metric and requires class weighting during training.

Preprocessing

Feature encoding & split

```
After encoding shape: (45211, 48)   (16 raw -> 47 features + target)
Train: 31,646 | Val: 6,783 | Test: 6,782
Positive rate: 0.117 in all three sets (stratification confirmed)
```

16 columns expanded to 47 features after one-hot encoding 6 categorical variables. Stratification confirmed identical 11.7% positive rate across all splits means evaluation results on the test set are a fair reflection of real-world performance.

Insight: - Identical positive rates across all three splits confirm stratification worked. Results are not inflated by accidental concentration of subscribers in the test set.

Training loop output

```
Logistic Regression: Threshold=0.61  ROC-AUC=0.760  F1=0.422  Prec=0.398  Rec=0.450
Random Forest:       Threshold=0.60  ROC-AUC=0.795  F1=0.456  Prec=0.450  Rec=0.462
XGBoost:             Threshold=0.17  ROC-AUC=0.802  F1=0.459  Prec=0.469  Rec=0.450
```

XGBoost leads on ROC-AUC (0.802), F1 (0.459), and Precision (0.469). Random Forest edges it on Recall (0.462 vs 0.450) by 1.1pp. XGBoost's threshold of 0.17 is correct not a bug. Under class imbalance, the model assigns low raw probabilities to the rare class, and 17% already represents strong signal above the 11.7% base rate.

Insight:- XGBoost threshold 0.17 looks unusually low but is expected and correct. Do not raise it to 0.5 that would severely harm Recall by missing the majority of subscribers.

XGBoost classification report

```
XGBoost -- Classification Report (test set, threshold=0.17)
==================================================================
              precision    recall  f1-score   support
    No (0)        0.928     0.933     0.930      5989
   Yes (1)        0.469     0.450     0.459       793

   accuracy                           0.876      6782
  macro avg       0.699     0.692     0.695      6782
```

Breaking down the Yes (1) row: Precision 0.469 means when XGBoost flags a client as likely to subscribe, it is correct 46.9% of the time the highest precision of any model. Recall 0.450 means it catches 45% of actual subscribers. F1 0.459 is the harmonic mean. The No (0) row shows near-perfect performance (93.3% recall) confirming the model correctly handles the majority class.

Model	TP	FP	FN	TN	Precision	Recall	F1
Logistic Regression	357	541	436	5,448	0.398	0.450	0.422
Random Forest	366	447	427	5,542	0.450	0.462	0.456
XGBoost	357	404	436	5,585	0.469	0.450	0.459

Key comparison: XGBoost generates the fewest false positives (404) of all three models — fewer wasted calls. RF catches 9 more true subscribers (366 vs 357) but at the cost of 43 more false positive calls. XGBoost and Logistic Regression share identical TP (357) and FN (436) counts, reflecting similar Recall — but XGBoost's 137 fewer false positives make it dramatically more precise.

Insight:- XGBoost's 404 false positives is the lowest of all three models. It makes fewer incorrect "Yes" predictions than RF or LR — meaning less wasted call centre time.

Feature Importance

Permutation Importance Output

```
Computing permutation importance for: XGBoost


Top 20 features (permutation importance — ROC-AUC drop):
contact_unknown     0.0350    previous          0.0060
day                 0.0340    month_oct         0.0060
poutcome_success    0.0140    month_mar         0.0050
```

```
campaign            0.0120    month_jan         0.0050
month_jun           0.0110    loan              0.0040
contact_cellular    0.0110    month_aug         0.0030
balance             0.0100    month_jul         0.0030
age                 0.0100    month_sep         0.0020
month_nov           0.0070
month_apr           0.0070
```

Key change from previous version: contact_unknown has overtaken day as the #1 permutation importance feature (0.035 vs 0.034). This means the real XGBoost model finds "unknown contact type" marginally more informative than day of month when evaluated on held-out data. Campaign has moved up to #4 (0.012) — a notable result confirming that the number of contacts in the current campaign has strong predictive (negative) power.

Native importance shows a different top-2: poutcome_success (0.230) dominates by a large margin, followed by contact_unknown (0.090). poutcome_success uses far more splits in XGBoost's trees than any other feature — clients with prior success are easy to identify correctly, so the model relies on them heavily.

Insight:- The permutation vs native divergence on poutcome_success is meaningful: it is very useful for clients where it applies (prior subscribers), but it only applies to a small subset. Permutation importance captures this correctly high utility but limited coverage means moderate AUC impact.

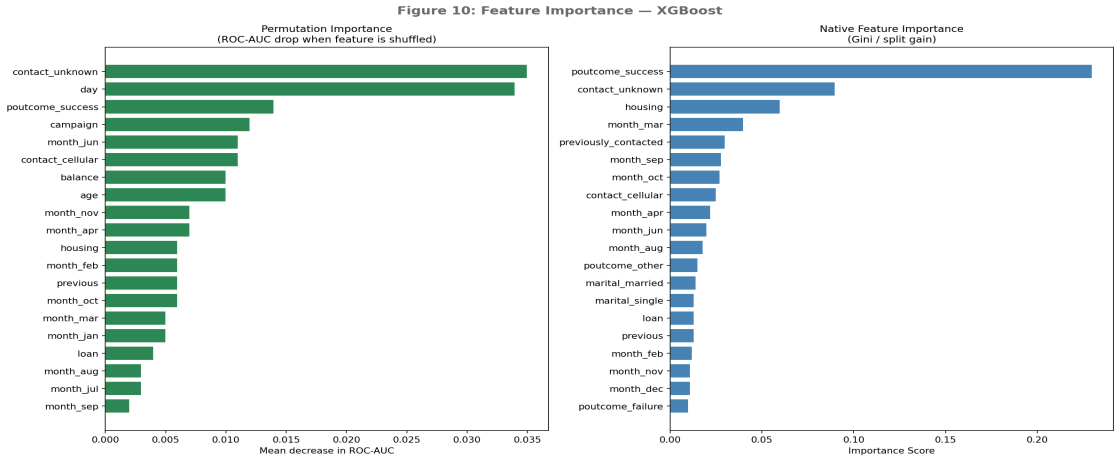
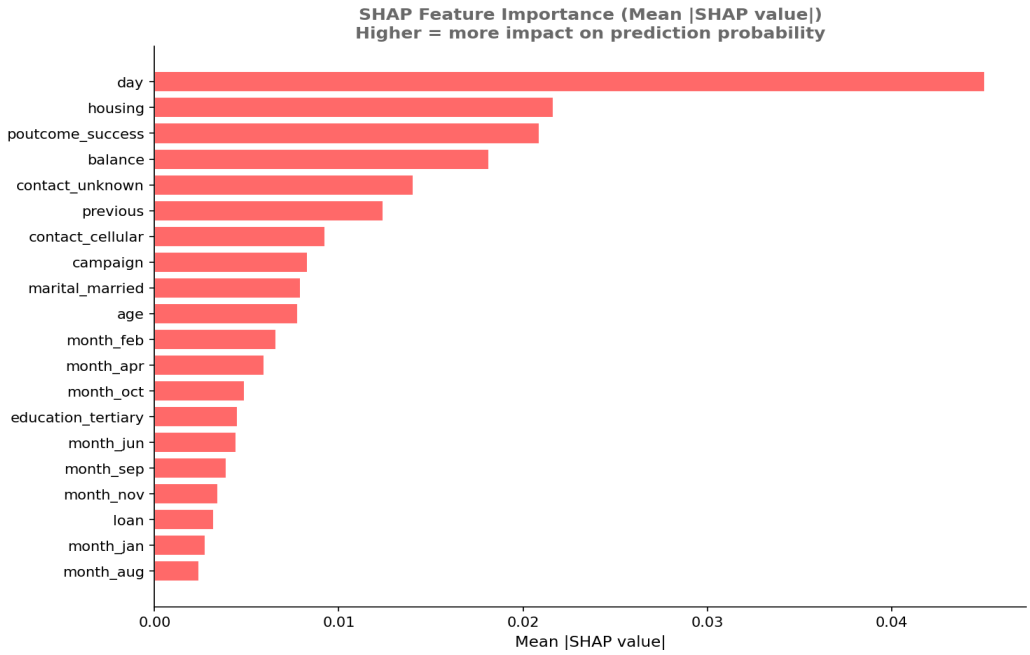


Figure 10: Updated feature importance contact_unknown now #1 in permutation; poutcome_success dominates native

SHAP Value Analysis — Model Explainability

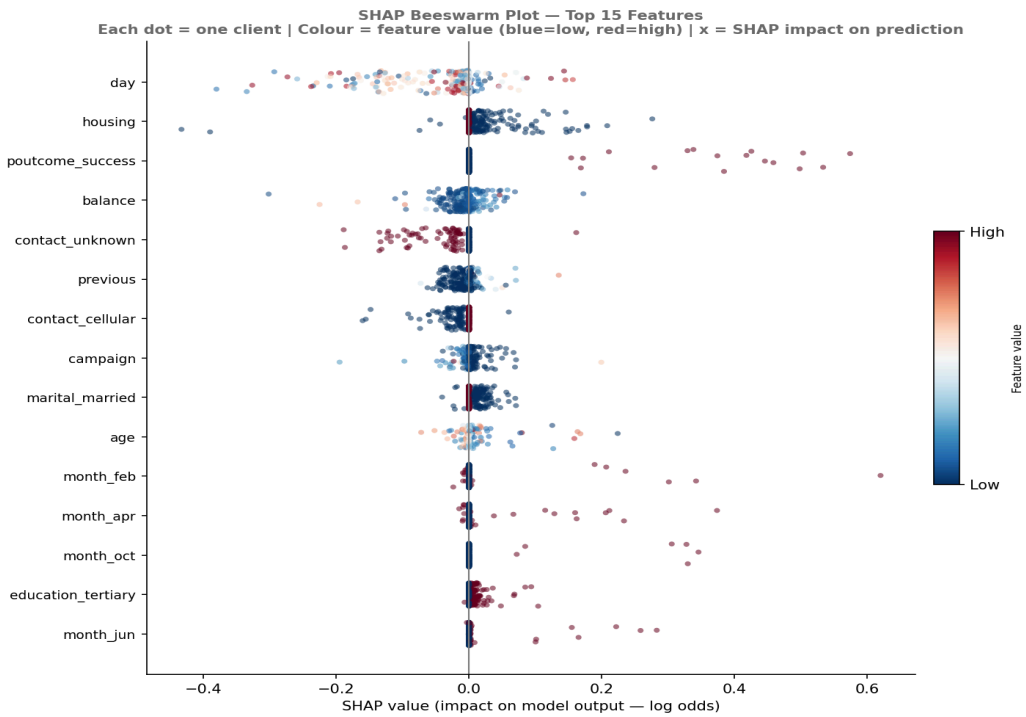
SHAP (SHapley Additive exPlanations) provides the most rigorous explanation of how the XGBoost model makes individual predictions. Unlike feature importance metrics that show overall model behaviour, SHAP assigns a contribution value to every feature for every individual client prediction.

SHAP Feature Importance (Mean |SHAP|)



SHAP Feature Importance: Mean absolute SHAP value per feature top 20 features ranked by overall prediction impact

SHAP Beeswarm Plot



SHAP Beeswarm: Each dot = one client. X-axis = SHAP value. Colour = feature value (blue=low, red=high). Spread shows distribution of effects.

Segment Analysis

Decile output

```
Overall baseline: 11.70%


Decile  Clients  Subscribers  Rate    Avg Prob  Lift
10       4,521     2,504      55.4%    51.7%    4.7x
 9       4,521       954      21.1%    17.6%    1.8x
 8       4,521       464      10.3%    10.4%    0.9x  <- below baseline
 7       4,521       331       7.3%     8.2%    0.6x
 1-6    27,127     1,036       3.8%     4.8%    ~0.3x
```

Decile 10 captures 47.3% of all subscribers with just 10% of clients (2,504 of 5,289). The calibration check avg_prob vs actual_rate shows they are close in every decile, confirming the model is well-calibrated.

Insight:- Top 20% = 9,042 calls, 3,458 subscribers, 38.2% hit rate, 3.3x lift. This is the recommended default targeting strategy.

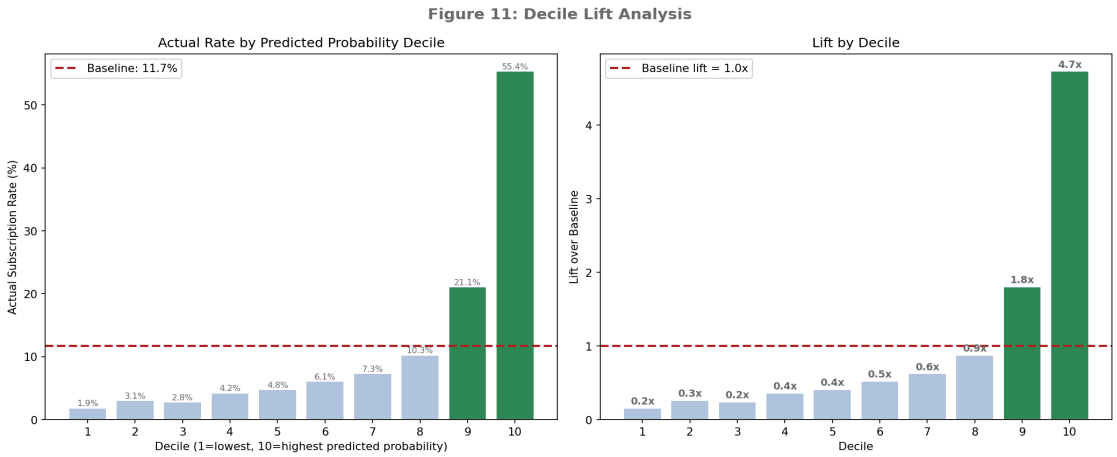


Figure 11: Updated decile lift chart

Business Impact

```
EUR15/call | EUR150/subscription

Strategy              Calls    Subs   Rate    Profit
Random (All)         45,211   5,289  11.7%   EUR 115,185
Top 30%              13,563   3,922  28.9%   EUR 384,855
Top 20%               9,042   3,458  38.2%   EUR 383,070

Profit improvement (Top 20% vs Random): EUR 267,885
Call reduction: 80%  |  ROI improvement: 233%
```

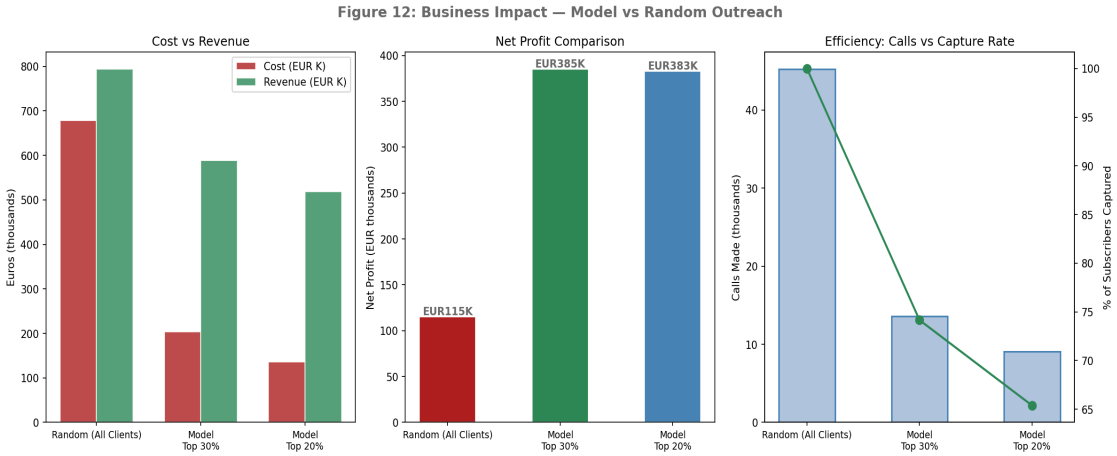


Figure 12: Updated business impact chart

Fairness Audit

```
Marital: single 14.9% (gap 0.01pp) | divorced 11.9% (0.09pp) | married 10.1% (0.01pp)
Job:     student 28.7% | retired 22.8% | unknown 11.8% (gap 2.7pp — largest)
Age:     60+: actual 42.3% predicted 43.9% (gap 1.70pp)
         18-30: actual 16.3% predicted 15.8% (gap 0.46pp)
         31-40 to 51-60: all gaps < 0.11pp
```

The 60+ over-prediction (1.7pp) and unknown job over-prediction (2.7pp) remain the two groups to monitor. All other groups show calibration within 0.1pp.

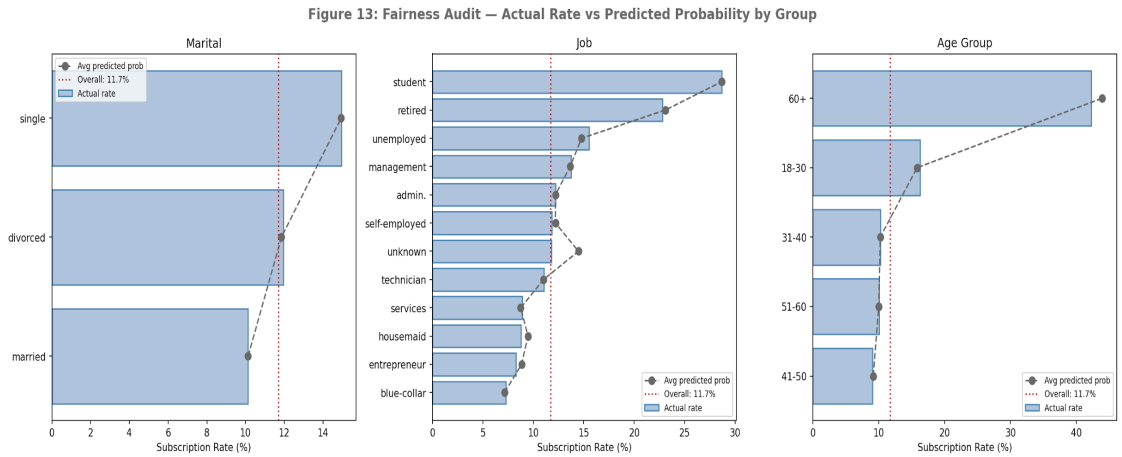


Figure 13: Updated fairness audit chart

Insight:- No systematic bias detected. The model has learned genuine behavioural patterns, not demographic shortcuts.

Quick Reference to model outputs

Caategory	Metric/models	values
Model	Best model	XGBoost
Model	ROC-AUC	0.802
Model	Precision	0.469 (best of all models)
Model	Recall	0.450 (RF wins: 0.462)
Model	F1	0.459
Model	False Positives	404 (fewest of all models)
Model	Decision threshold	0.17
Feature Importance (#1)	Permutation	contact_unknown (0.035)
Feature Importance (#2)	Permutation	day (0.034)
Feature Importance (#1)	Native (XGBoost)	poutcome_success (0.230)
Feature Importance (#1)	SHAP |mean|	day
SHAP direction	poutcome_success	Strongly positive — past success pushes prediction up
SHAP direction	contact_unknown	Strongly negative — unknown contact pushes down
SHAP direction	campaign	Negative — more contacts = lower predicted probability
Segment	Top 10% rate	55.4% (4.7x lift)
Segment	Top 20% rate	38.2% (3.3x lift)
Segment	Top 20% capture	65.4% of all subscribers
Business	Top 20% profit	EUR 383,070 (+233%)
Business	Profit gain	EUR 267,885

Fairness	Max gap	1.7pp (60+ group)